YEAR 8 MATHS · CHAPTER 9 · MIND MAP

Linear relationships

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1 The Cartesian plane

(x, y): across first, then up or down. The origin is $(0, 0)$.

$(-2, 3)$ is 2 left and 3 up, in the second quadrant

2 Rules, tables and graphs

Use the rule to fill a table, plot the points and join them.

$y = 2x + 1$: $x = 0, 1, 2$ gives $y = 1, 3, 5$

3 Finding the rule

The step in y is the number in front of x. The value at $x = 0$ is the constant.

x: 0, 1, 2, 3 and y: 4, 7, 10, 13

Up 3 each time, starts at 4: $y = 3x + 4$

4 Solving equations with graphs

Read across from the value you want to the line, then down to the x-axis.

Solve $2x + 1 = 7$ on $y = 2x + 1$: at $y = 7$, $x = 3$

5 Solving inequalities with graphs

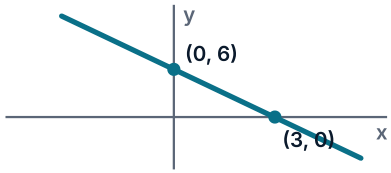
Find the part of the line above or below the value you want.

$2x + 1 < 7$: the line is below $y = 7$ when $x < 3$

6 The x- and y-intercepts

x-intercept: put $y = 0$. y-intercept: put $x = 0$.

$2x + y = 6$: x-intercept $(3, 0)$, y-intercept $(0, 6)$



7 Gradient

Gradient $= \frac{\text{rise}}{\text{run}}$. Up to the right is positive, down is negative.

$(1, 2)$ to $(3, 6)$: $\frac{4}{2} = 2$

Horizontal lines have a gradient of 0

8 Gradient–intercept form

$y = mx + c$: m is the gradient, c is the y-intercept.

$y = 3x - 2$: $m = 3,\ c = -2$

$y = -x + 4$: $m = -1,\ c = 4$

9 Applications of straight lines

c is the starting value, m is the rate.

Taxi: \$3 plus \$2 per km: $C = 2d + 3$

5 km: $C = 2 \times 5 + 3 = \$13$

10 Non-linear graphs

Not every rule gives a straight line. $y = x^2$ is a parabola.

$y = x^2$: $(-2, 4), (0, 0), (2, 4)$

